Data Flow Diagram

Date	11 July 2026
Team ID	XXXXXX
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

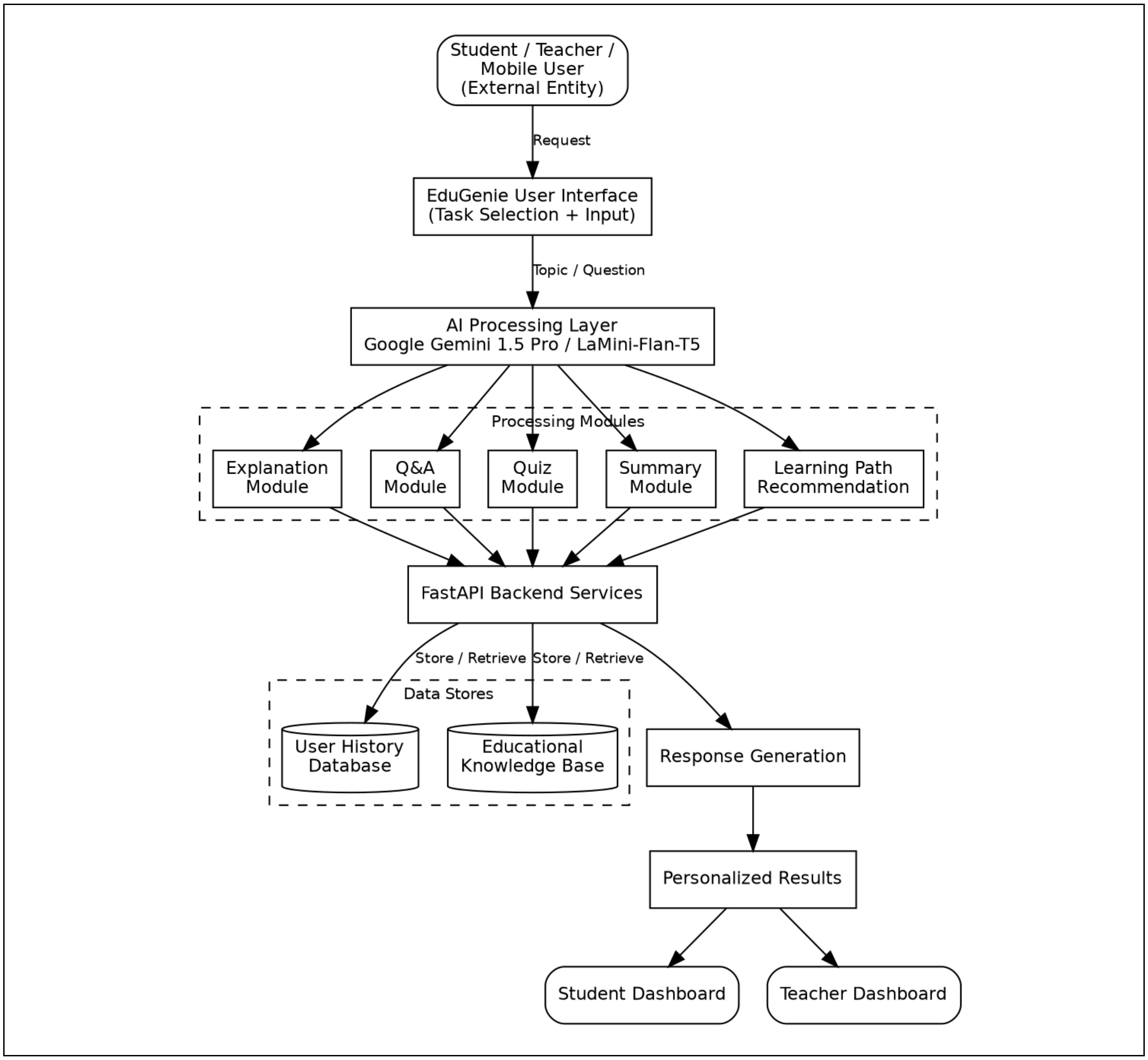
Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how information moves through EduGenie – Google Gemini Powered Learning Assistant. It represents the interaction between the student, AI models, system processes, data stores, and the flow of information from user input to AI-generated educational responses. The diagram provides a high-level overview of how user queries are processed, analyzed, and transformed into personalized learning outputs.

DFD Symbol Legend

Symbol	Name	Description
	External Entity	Represents the student who submits learning requests and receives AI-generated educational responses.
	Process	Represents processing activities such as Question Answering, Concept Explanation, Quiz Generation, Text Summarization, Learning Path Recommendation, and AI Response Generation.
	Data Store	Stores user queries, generated quizzes, summaries, learning recommendations, and application logs for future reference.
	Data Flow	Indicates the movement of information between the student, AI modules, processes, and data stores.

Diagram



Data Flow Diagram Description

The workflow begins when the student opens the EduGenie application and selects a learning task such as Question Answering, Concept Explanation, Quiz Generation, Text Summarization, or Personalized Learning Path. The student enters a topic or academic question through the user interface. The request is forwarded to the FastAPI Backend, which processes it using Google Gemini 1.5 Pro and LaMini-Flan-T5 AI models.

The AI models analyze the request and generate the appropriate educational response based on the selected task. The generated explanations, answers, quizzes, summaries, or learning recommendations are temporarily stored in the application database or cache. Finally, the processed results are displayed through the EduGenie user interface, allowing the student to review the output, continue learning, and improve academic performance.